

API Summary

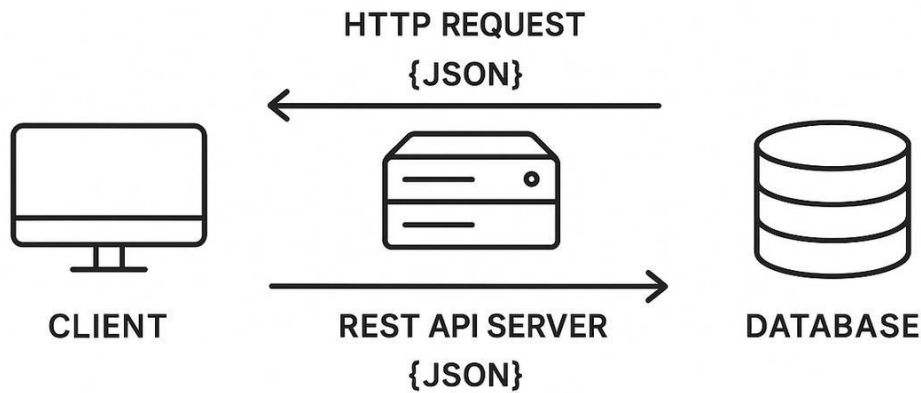
Project: Meal Planning and Recipe Management Web Application

1. Overview

The application communicates with a backend developed using Flask through RESTful API endpoints. These endpoints manage data operations related to recipes, meal plans, and nutritional calculations. All data is exchanged in JSON format and stored in a MongoDB database.

Angular utilises the HTTPClient module to send asynchronous HTTP requests and dynamically update the user interface based on API responses.

2. API Communication Flow



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Figure 1: API Communication Flow

The diagram above illustrates the communication process between the frontend and backend. When a user performs an action, such as creating or retrieving a recipe, the Angular frontend sends an HTTP request using the HTTPClient service. The Flask backend processes the request, interacts with the MongoDB database, and returns a JSON response. The frontend then dynamically updates the user interface based on the returned data.

3. API Endpoints

3.1 Recipe Endpoints

GET /api/recipes

- Retrieves all recipes from the database
- Used to display recipe lists

GET /api/recipes/{id}

- Retrieves a specific recipe by its unique ID

POST /api/recipes

- Creates a new recipe

Request Body Includes:

- title
- ingredients
- instructions (stored as an array)
- calories
- protein
- carbohydrates
- servings
- image URL

PUT /api/recipes/{id}

- Updates an existing recipe

DELETE /api/recipes/{id}

- Deletes a recipe from the system

3.2 Meal Plan Endpoints

GET /api/mealplans

- Retrieves all meal plans

POST /api/mealplans

- Creates a new meal plan

PUT /api/mealplans/{id}

- Updates a meal plan

DELETE /api/mealplans/{id}

- Deletes a meal plan

3.3 Nutrition Endpoint

POST /api/nutrition/calculate

- Calculates daily calorie requirements and macronutrient distribution based on user input

Inputs:

- Age
- Weight
- Height
- Gender

Outputs:

- Daily calorie requirement
- Protein intake
- Carbohydrates
- Fat intake

4. Data Handling

- Data is stored in MongoDB collections as JSON documents
- Recipe instructions are stored as arrays for structured display
- Image URLs are used instead of file uploads to reduce server load
- API responses are returned in JSON format and processed in Angular

5. Changes Since CW1

The following enhancements were implemented compared to the previous coursework:

- Introduction of meal plan management endpoints
- Addition of nutrition calculation endpoint
- Modification of recipe data structure to support instruction arrays
- Inclusion of nutritional attributes (calories, protein, carbohydrates, servings)
- Implementation of role-based access logic for admin and user roles

6. Error Handling

- Invalid requests return appropriate HTTP status codes (e.g., 400, 404)
- Input validation ensures required fields are provided
- Errors are handled in Angular and displayed to users for improved usability

7. Integration with Frontend

- Angular services use HTTPClient for API communication
- Observables manage asynchronous data flow
- Responses are dynamically rendered in Angular components
- UI updates occur without page reloads

8. Summary

The API provides a structured and scalable mechanism for handling application data. It enables efficient communication between the frontend and backend while supporting all required functionalities, including CRUD operations and advanced nutritional calculations.